

Unit IV

ETL Process

4.1 Introduction to ETL Process

4.2 Data Extraction – Identification of Data Sources, Types –
Immediate Data Extraction and Deferred Data Extraction

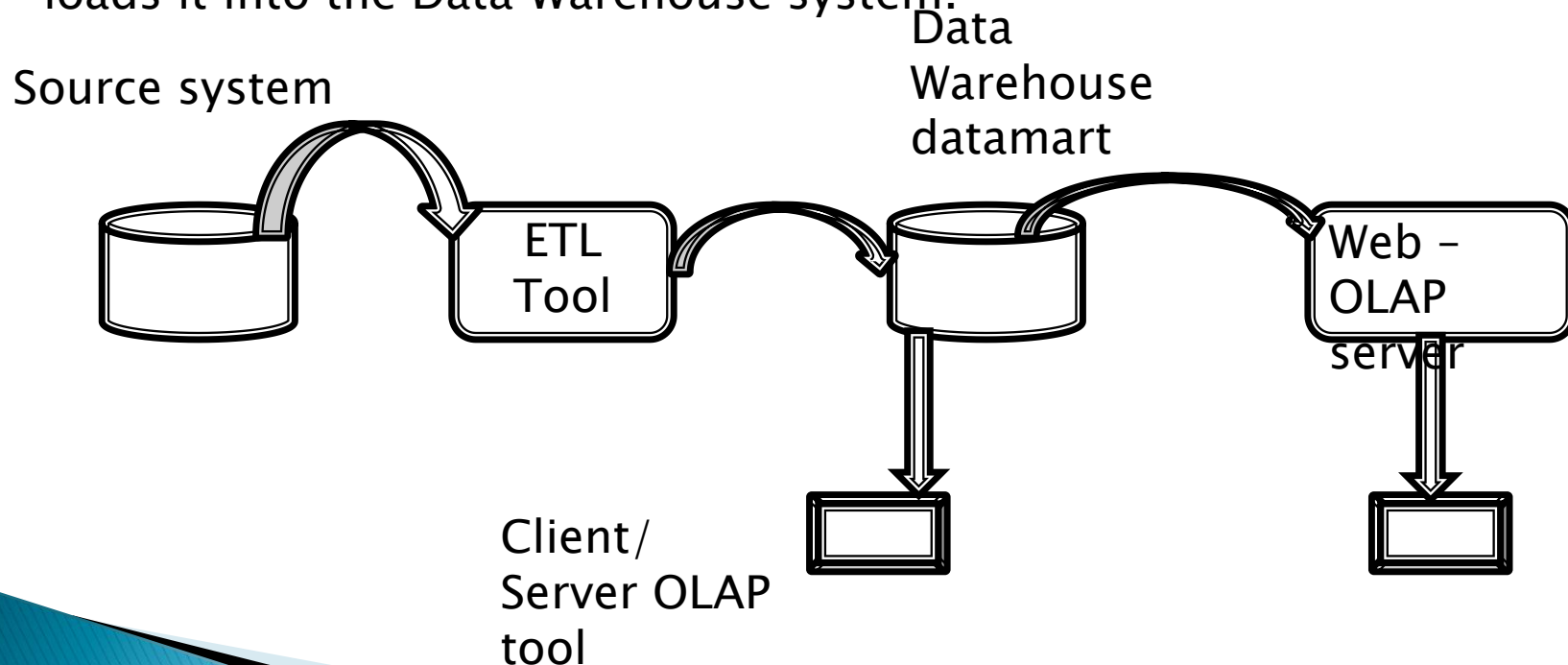
4.3 Data Transformation: Tasks Involved in Data
Transformation

4.4 Data Loading: Techniques, Loading the Fact Tables and
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Introduction to ETL Process

ETL is a process in Data Warehousing and it stands for Extract, Transform and Load. It is a process in which an ETL tool extracts the data from various data source systems, transforms it in the staging area, and then finally, loads it into the Data Warehouse system.

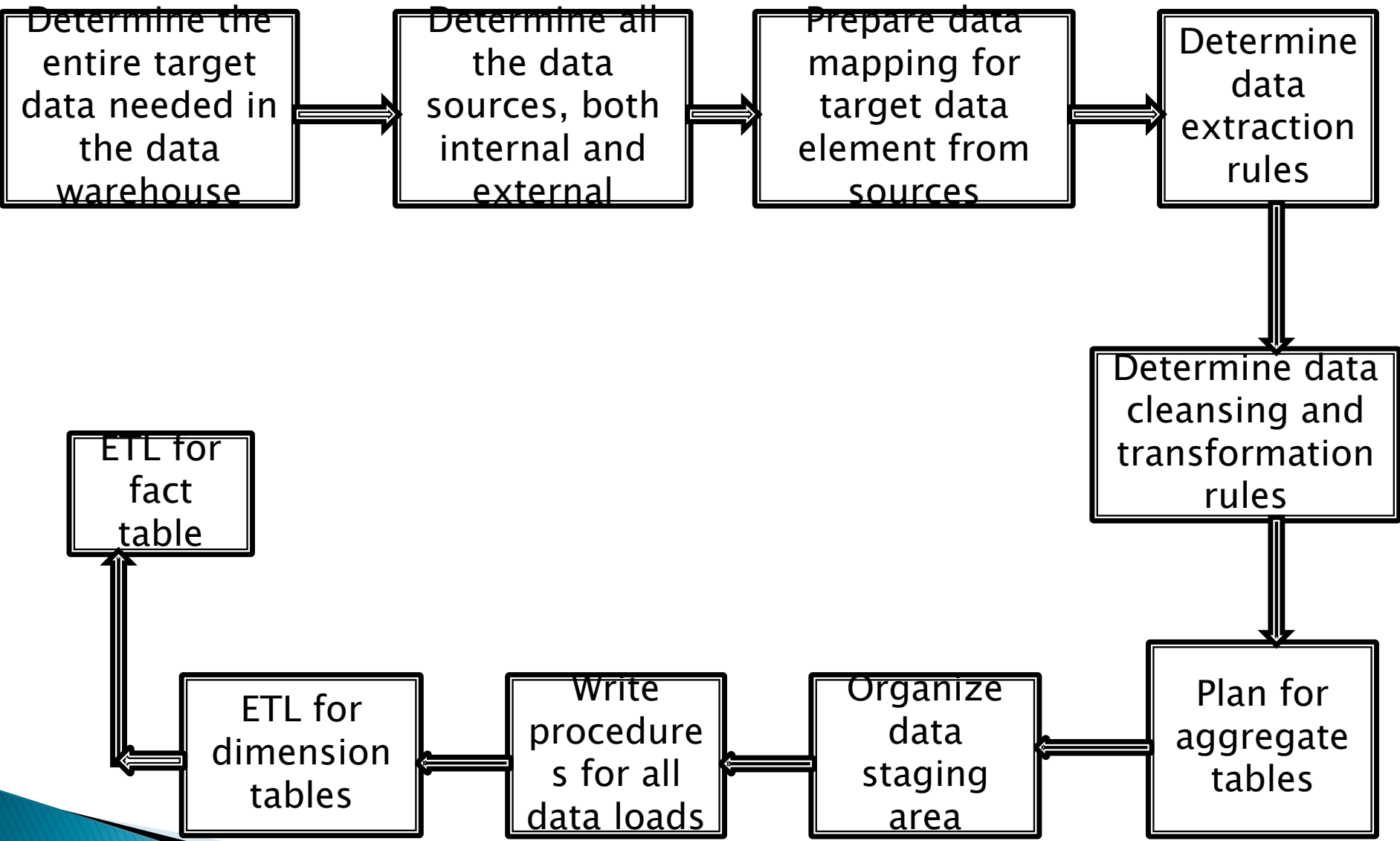


Data transfer from source systems to the data warehouse

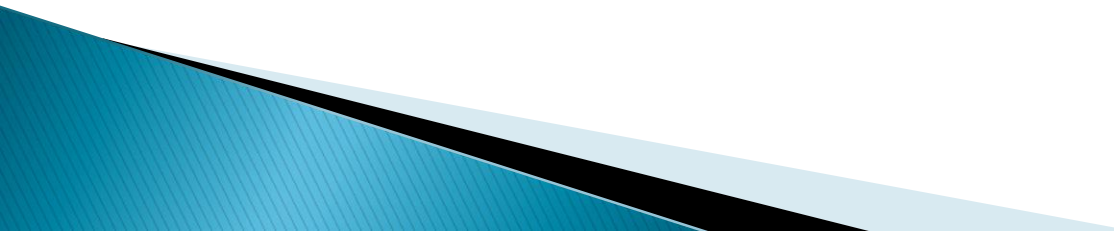
Challenges in ETL Process

Types of difficulties in the ETL functions:

- Source systems are diverse and disparate.
 - Source systems run on platforms and have different operating systems installed.
 - Most of the operational systems do not preserve historical data which is critical for a data warehouse.
 - Quality of data cannot be guaranteed in the older operational source systems.
 - Structure of the source systems keep changing over time with the advent of new technology.
- 



Major steps in the ETL

- The prevalence of data inconsistency in the source system, (i.e. same data element represented differently in different source systems) is a major challenge in the ETL functions.
 - Lack of consistency is one problem and the lack of means for resolving the inconsistencies is another problem.
 - Data in the source system may be ambiguous, or stored in a cryptic form which hardly provides any help to the users.
 - The data type, format, and naming convention may be different in different source systems.
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DATA EXTRACTION

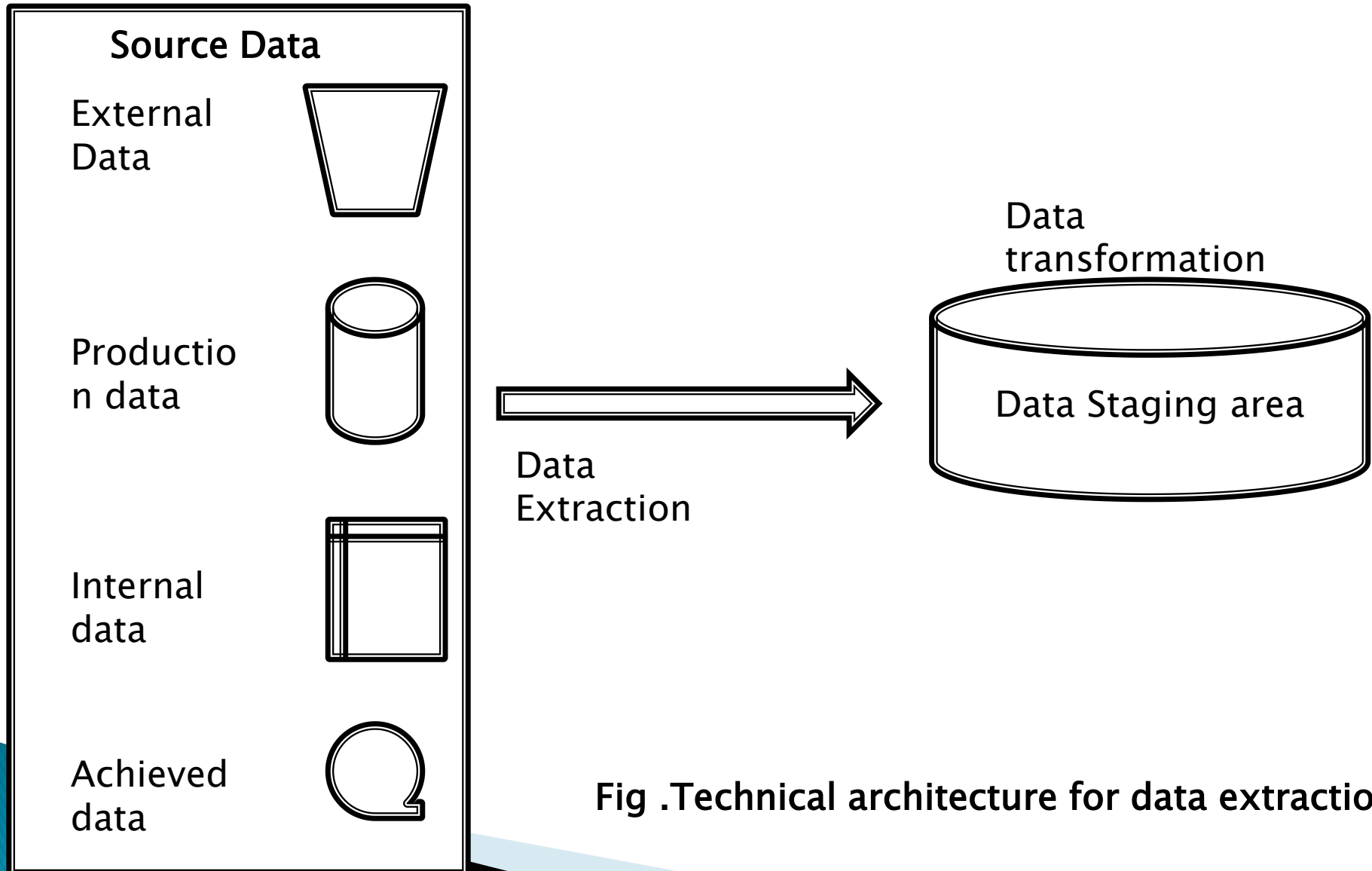


Fig .Technical architecture for data extraction

Issues And Formulate A Data Extraction Strategy For The Success Of The Data Warehouse

- Identify the applications and systems from which the data will be extracted.
- For each identified data source, determine the method for data extraction, i.e. Whether; it will be done manually or by using the tools.
- For each data source, determine the extraction frequency, i.e. decide whether the data will be extracted daily, weekly or monthly from the source systems.
- Estimate the acceptable time window for the extraction process from each data source.
- Determine the job sequencing feature that is, if the beginning of one extraction job has to wait until the previous one has completed or not.

Identification of Data sources

The process of source identification needs thoroughness, time, and exhaustive analysis.

Steps to identification of data sources –

- List every fact needed for analysis in the fact tables.
- For every dimension table, list each and every attribute.
- For each target data item, find the source system and the appropriate source data item.
- If there are multiple sources for the same data then choose the preferred sources.
- Formulate a consolidation rule for every data item that has multiple sources.
- Formulating splitting rules for every source field that will be distributed to multiple fields.
- Determine the default values.
- Search the source data for the missing values.

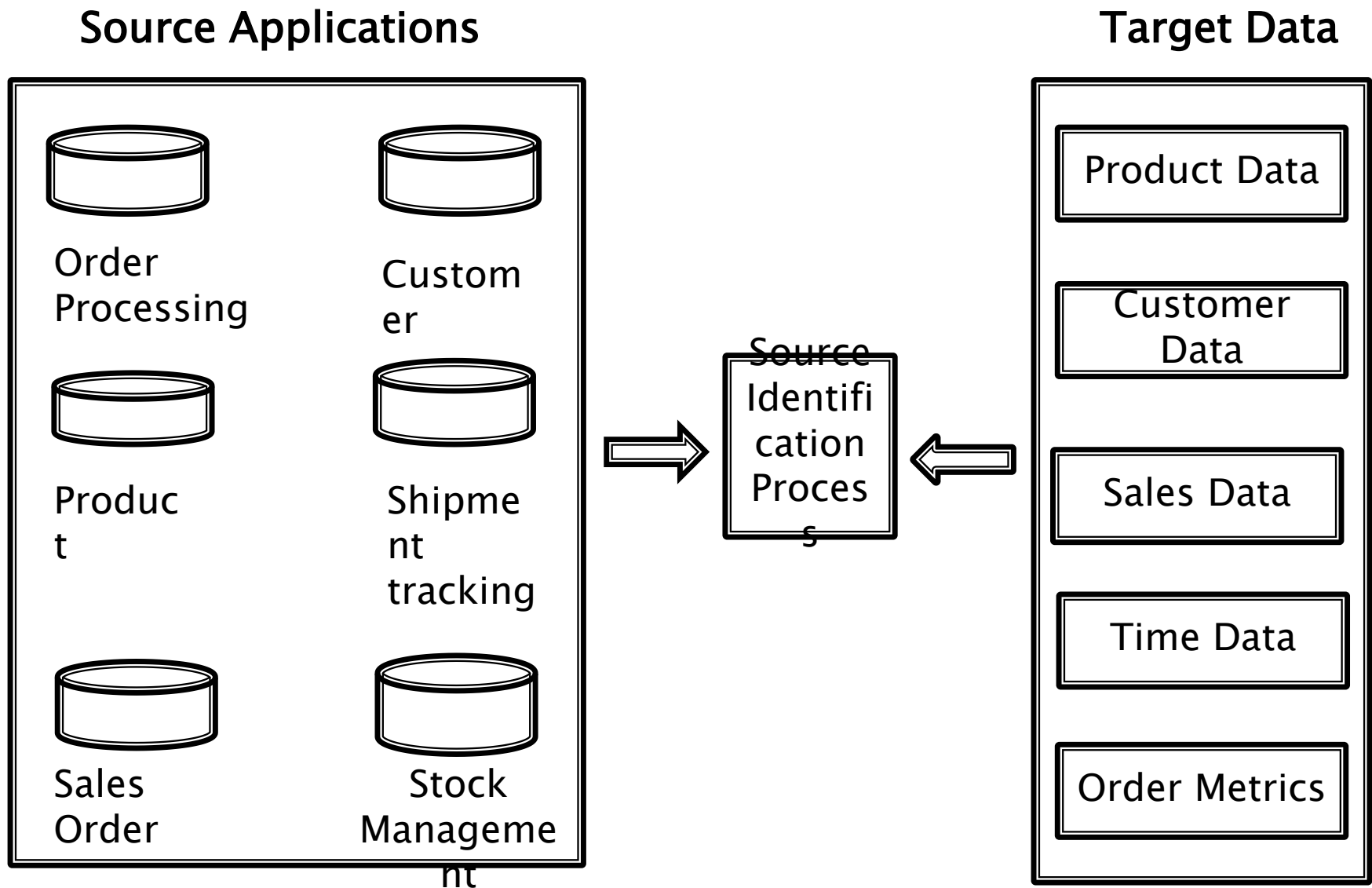


Fig : Identification of source system

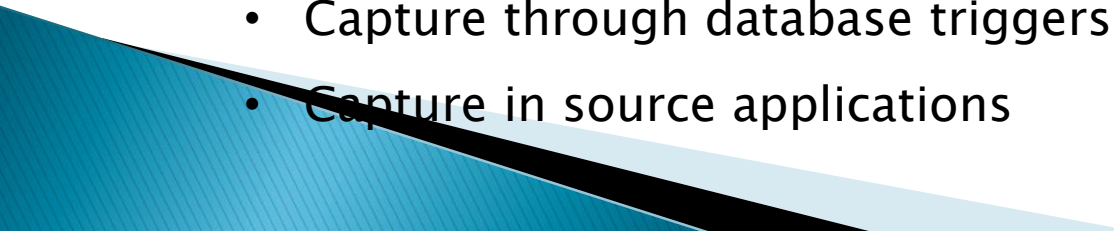
Extracting Data for Refreshing

Data extraction techniques are broadly classified into two categories –

- Immediate data extraction
- Deferred data extraction

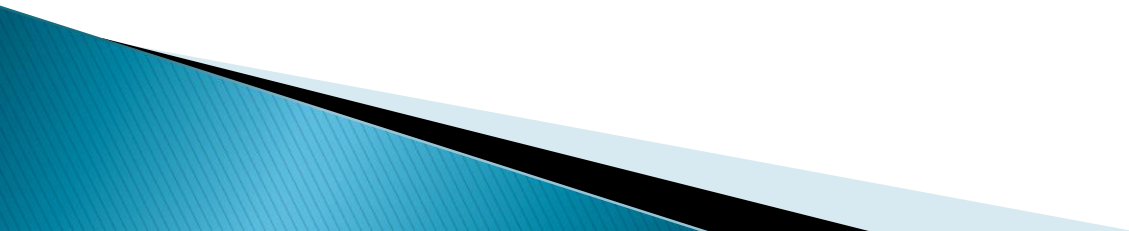
Immediate Data Extraction Technique

In this technique , the data extraction is real –time. It occurs as the transactions happen at the source databases and files. The three options for immediate data are :

- Capture through transaction logs
 - Capture through database triggers
 - Capture in source applications
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Capture through transaction logs

Capture through transaction logs technique reads the transaction log and selects all the committed transactions. The transaction logs are maintained for recovery from



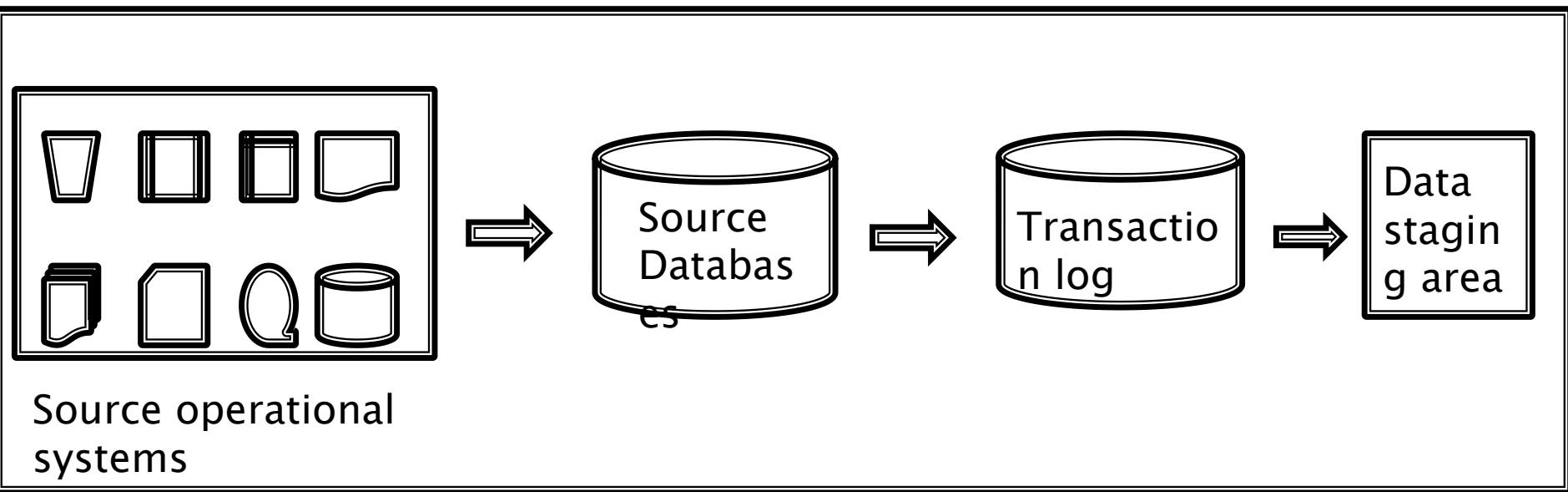
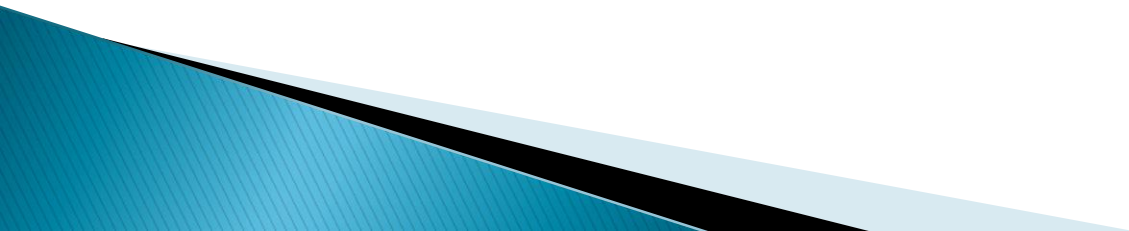


Fig. Capture through transaction logs

Capture through database triggers



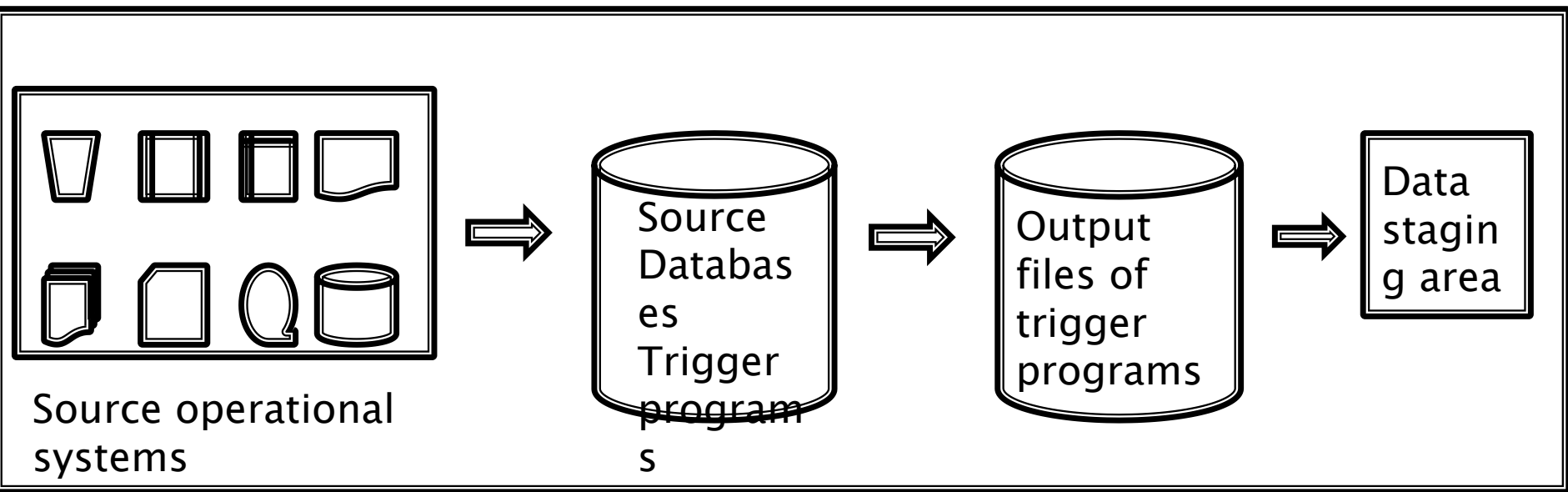
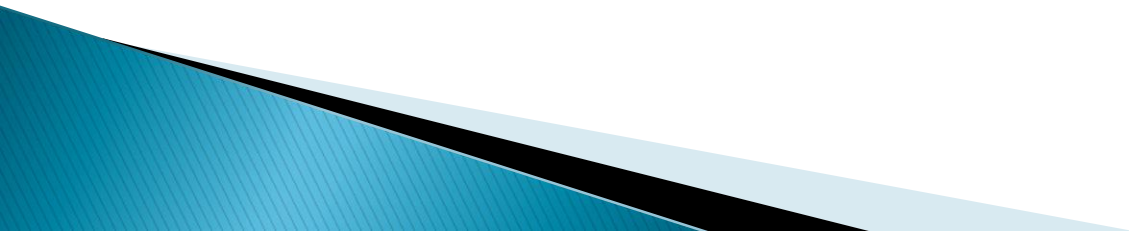


Fig. Capture through database triggers

Capture in source applications



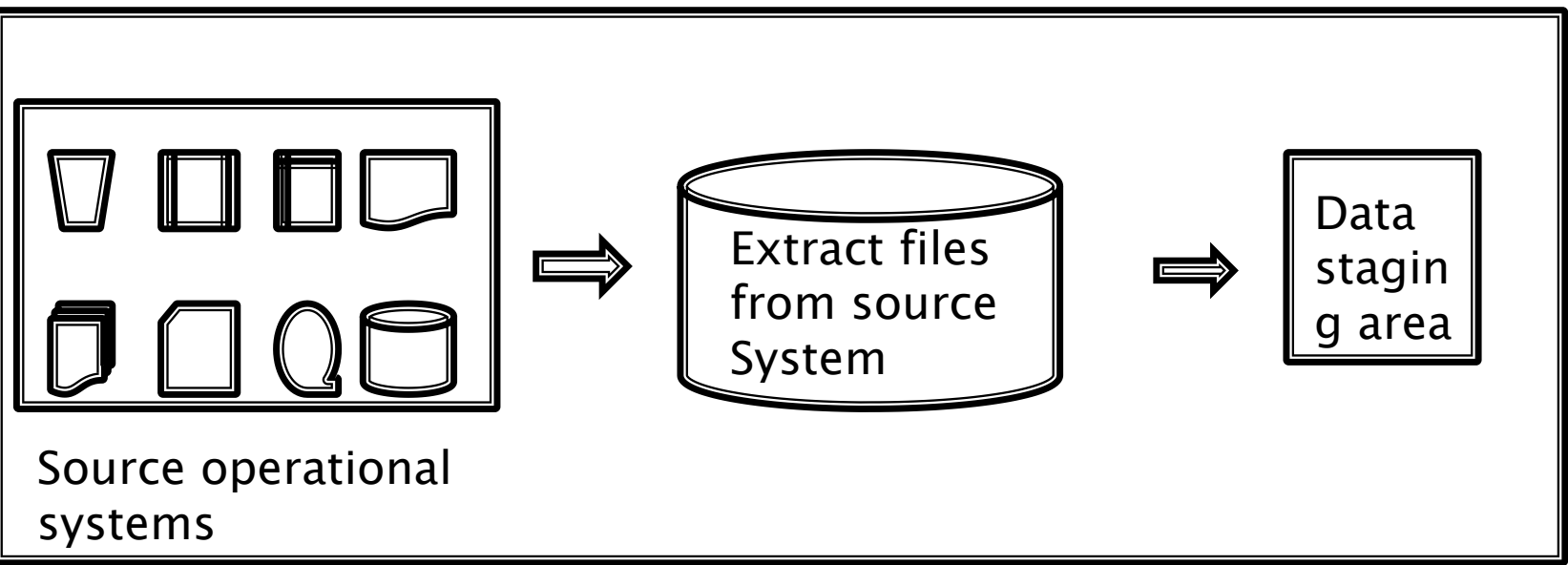


Fig. Capture in source applications

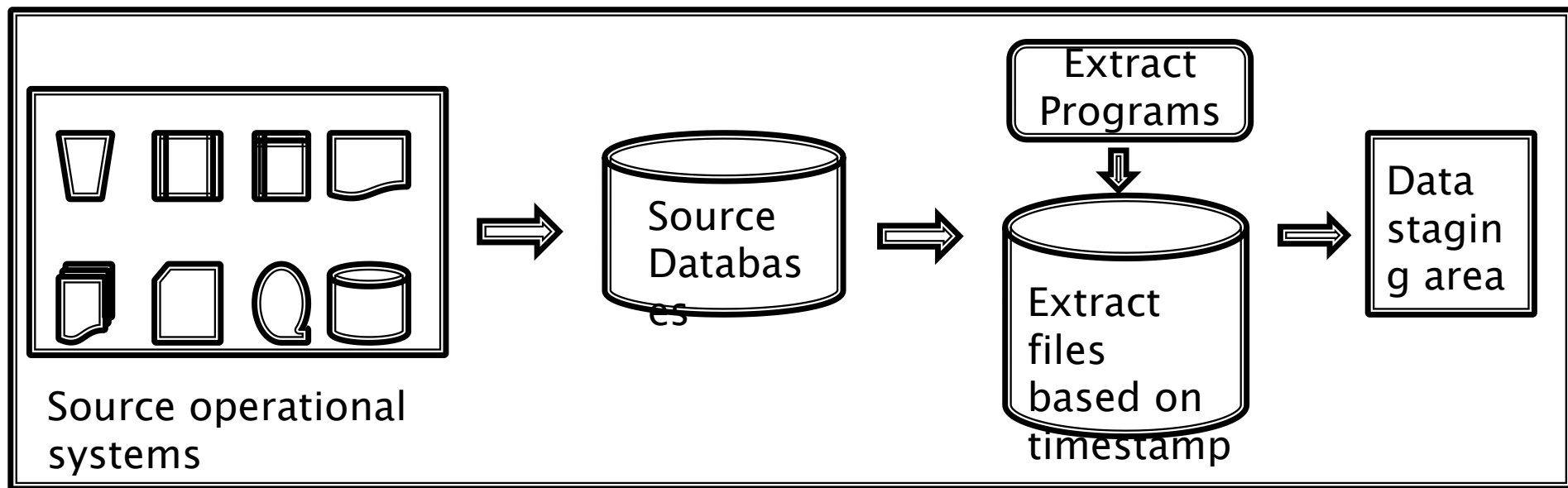
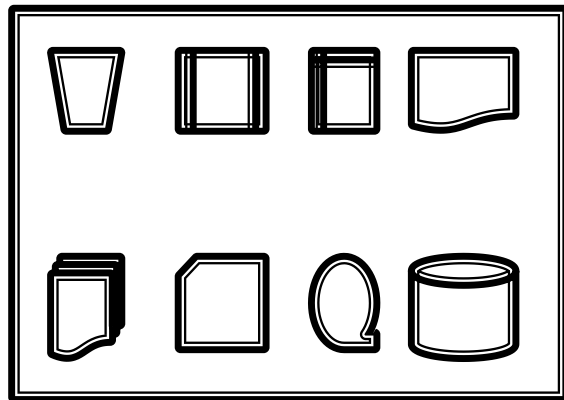


Fig. Capture based on data and timestamps



Source operational
systems

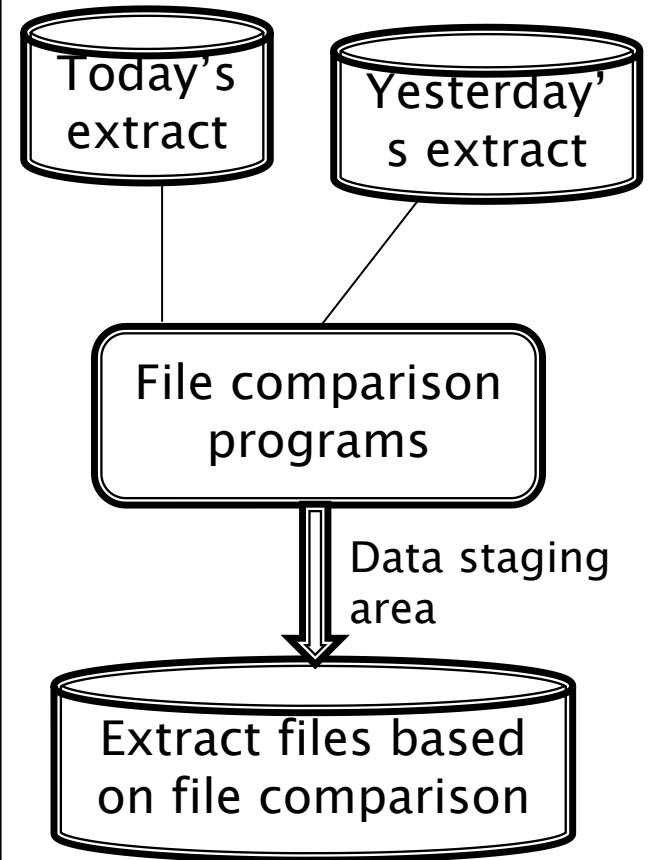
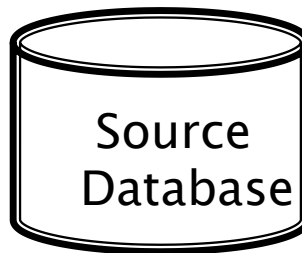


Fig. Capture by comparing files

Evaluation of extraction techniques

Capture through transaction logs	• Does not provide much flexibility for capturing specifications. • Does not affect the performance of source systems. • Does not require any revisions to the existing source applications. • Cannot be used on file-oriented system.
Capture through database triggers	• Does not provide much flexibility for capturing specifications. • Does not affect the performance of source systems. • Does not require any revisions to the existing source applications. • Cannot be used on legacy systems. • Cannot be used on file-oriented system.
Capture in	• Provides flexibility for capturing specifications.

Capture based on date and timestamp	• Provides flexibility for capturing specifications. • Does not affect the performance of source systems. • Requires the existing source applications to be revised. • Can be used on file-oriented system. • Cannot be used on legacy systems.
Capture by comparing files	• Provides flexibility for capturing specifications. • Does not affect the performance of source systems. • Does not requires the existing source applications to be revised. • May be used on file-oriented system. • May be used on legacy systems.

Data Transformation: Tasks Involved in Data Transformation

Tasks involved in Data Transformation

- Format Revision
 - Decoding of Fields
 - Splitting Fields
 - Merging of information
 - Character set conversion
 - Conversions of units
 - Date and time conversion
 - Summarization
 - Key restructure
 - De-duplication
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Format Revision

It include changes to the data types and lengths of individual data fields. Character set conversion, unit of measurement conversion, date/time conversion, etc

Decoding of Fields

Mapping NULL to 0 or "Male" to "M" and "Female" to "F," date format consistency, etc.

Splitting Fields

Splitting a single column into multiple columns

Merging of information

Bringing together relevant information from different data sources.
Ex- Customer name and code can be fetch from one table, his income level and age from another table and address and living style from yet another table.

Character set conversion

This type of data transformation is done to the textual data to convert its character set to an agreed standard character set.

Ex : – Convert the data from one character set to the other(ASCII)

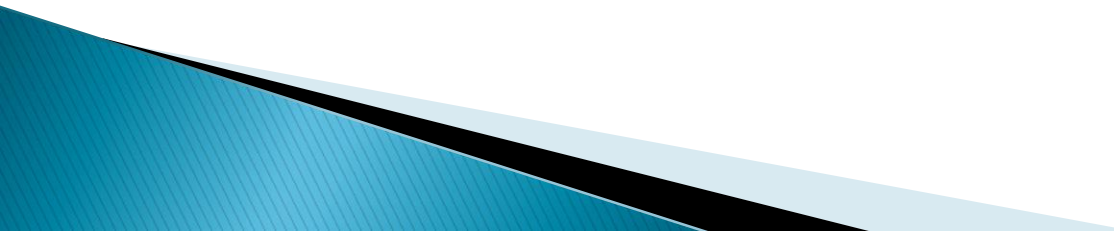
Conversions of units

Many company have global branches. So sales amount may be represented in different currencies in different source systems. Before entering data in data warehouse , you need to convert the figures into a common unit of measurement.

Date and time conversion

The data and time values also need to be represented in a standard format.

Ex: 1 October 2006 is written as 10/01/2006 in the US format
 01/10/2006 in the British Format



Summarization

Values are summarized to obtain total figures which are calculated and stored at multiple levels as business metrics – for example, adding up all purchases a customer has made to build a customer lifetime value (CLV) metric

De-duplication:

Identifying and removing duplicate records

Ex – COBOL and C lang , ETL software is of two type one produce code and another produces a parameterize run time model.

Key restructure

Establishing key relationships across tables

Operational system
key

Product Key Code

91	W1	R39	5678	07
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Data Loading

Initial Load: For the very first time loading all the data warehouse tables.

Incremental Load: Periodically applying ongoing changes as per the requirement. After the data is loaded into the data warehouse database, verify the referential integrity between the dimensions and the fact tables to ensure that all records belong to the appropriate records in the other tables. The DBA must verify that each record in the fact table is related to one record in each dimension table that will be used in combination with that fact table.

Full Refresh: Deleting the contents of a table and reloading it with fresh data.



Data loading challenges

Numerous ETL solutions are cloud-based, which is responsible for their speed and scalability. But large enterprises with traditional, on-premises infrastructure and data management processes often use custom-built scripts to collect and load their data into storage systems through customized configurations.

Slow down analysis: Every time a data source is added or changed, the system has to be reconfigured, which is time-consuming and hinders the ability to make quick decisions..

Increase the likelihood of errors: Changes and reconfigurations open the door to human error, duplicate or missing data, and other problems.

Require specialized knowledge: In-house IT teams often lack the necessary skills (and bandwidth) to code and monitor ETL tasks.

Require costly equipment: In addition to investing in the right human resources, organizations have to procure, house and maintain the hardware and other equipment to drive the process on site.



Techniques for Data Loading

Load : If the target table to be loaded already exists and contains some data records, then the load process will wipe out the existing data and store the data from the incoming file.

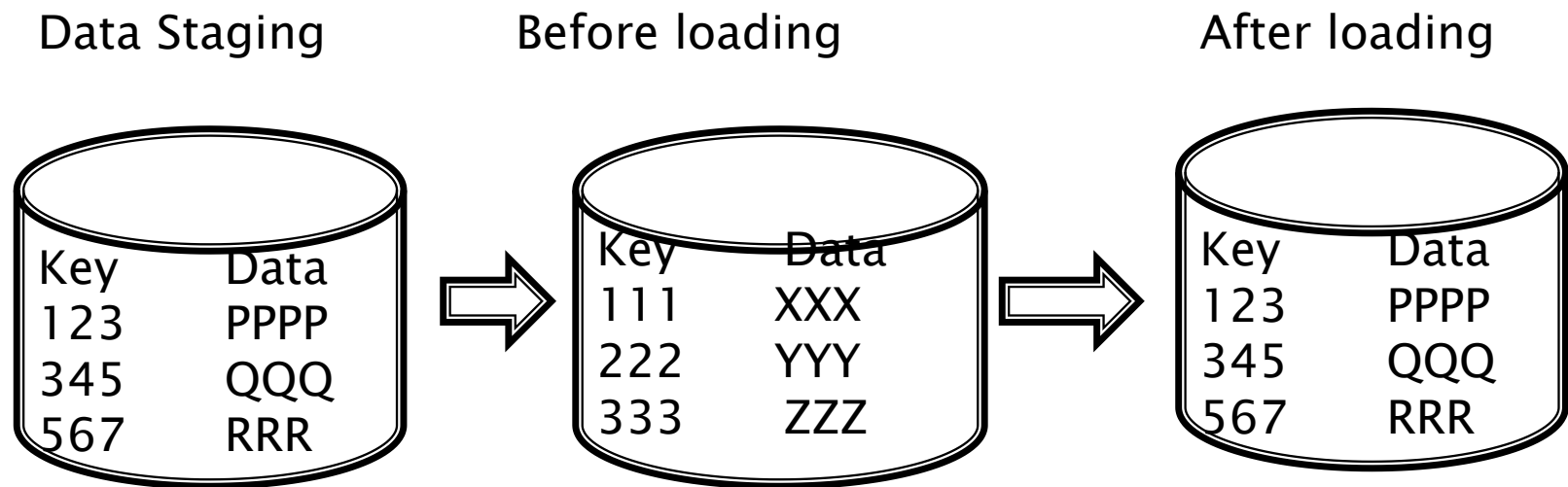
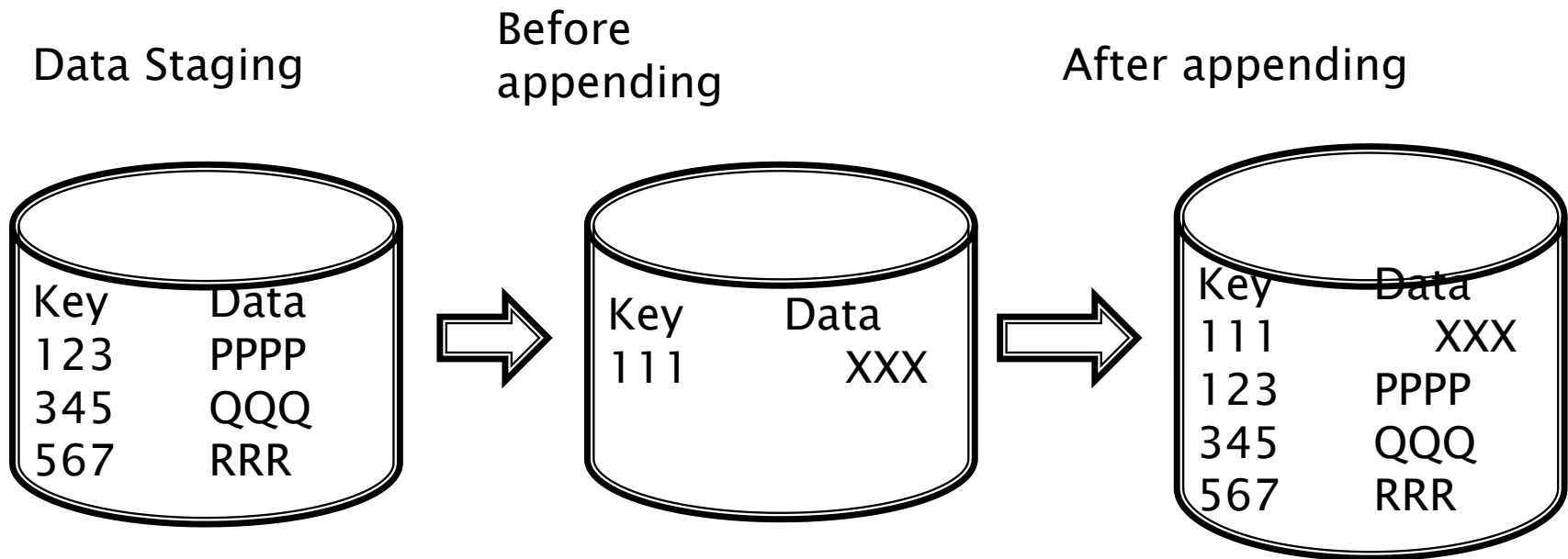


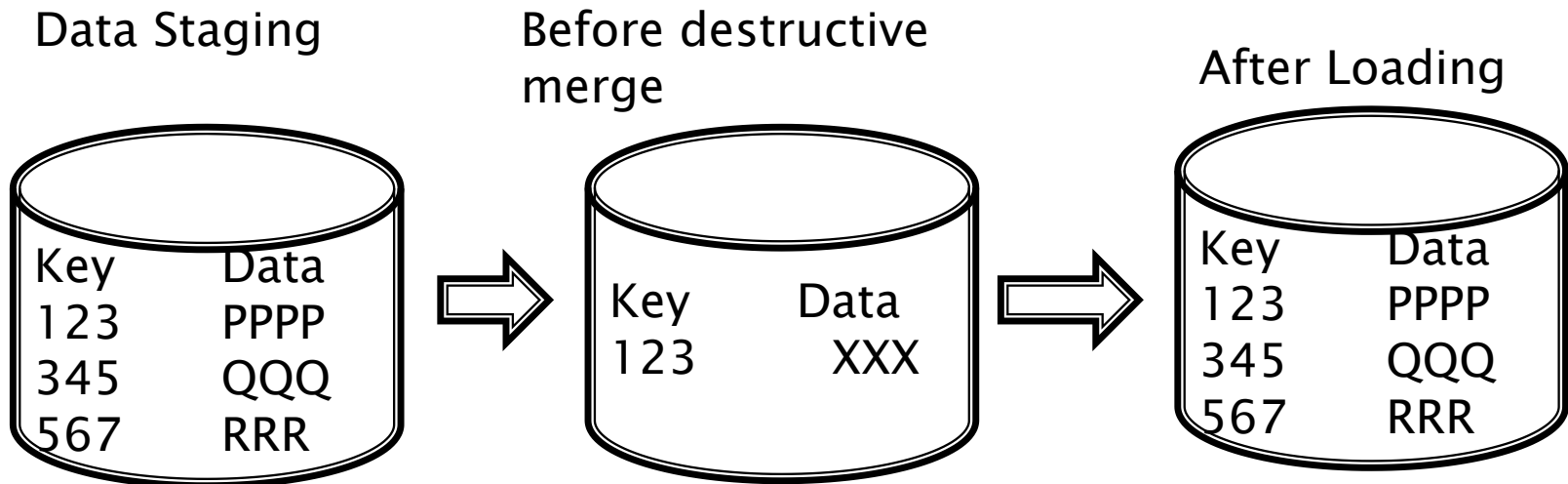
Fig : Load mode of data loading

Append: This mode is an extension to the previous one. If data already exists in the table, the append process adds the incoming data, thereby preserving the existing data in the target table.



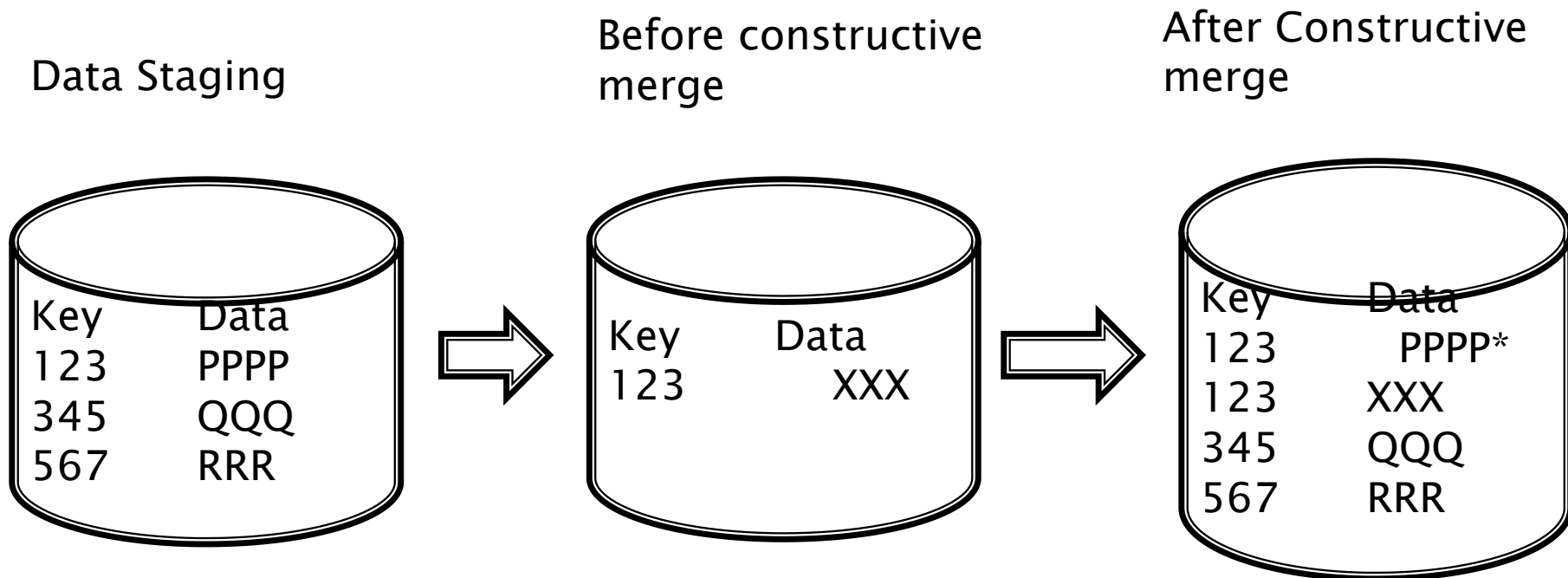
Destructive Merge:

In this mode, the incoming data is applied to the target data. If the primary key of the incoming record matches with the key of an existing record, then the existing record is updated. And in case the incoming record is a new record with an existing record, the incoming record to the target table.



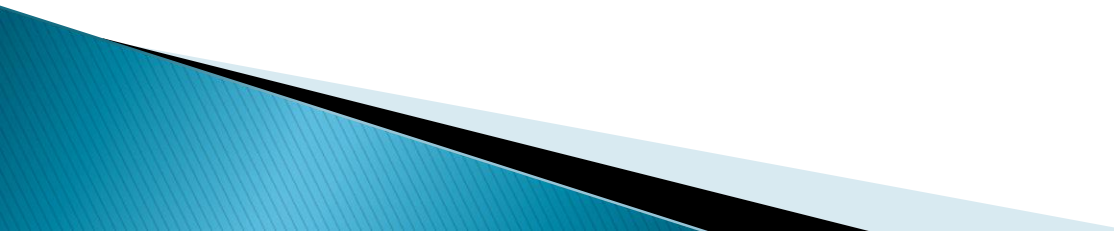
Constructive Merge:

If the primary key of an incoming record matches with the key on an existing record, then leave the existing record, add the incoming record and mark the added record as superceding the old record.



These modes of applying data to the data warehouse fit into the three types of loads.

Initial Load :

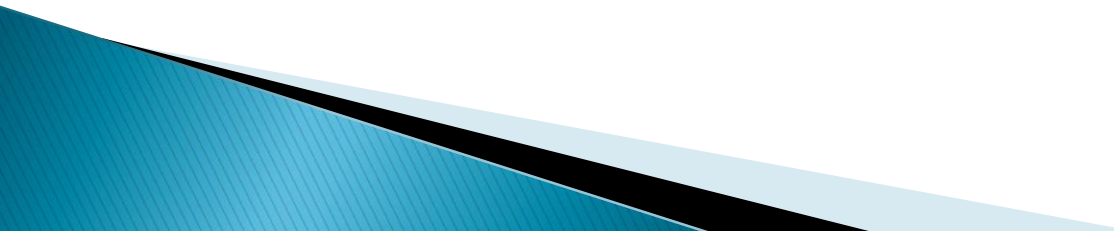
1. You can either load the whole data warehouse in a single run or split the load into separate subloads and run each of these subloads as single loads. Every load run create the database tables from scratch.
 2. If the initial load process takes more than one run to create a single table, then the load process has to be scheduled to run several days.
 3. Creation of indexing on initial loads or full refreshes needs special attention.
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Refresh versus Update

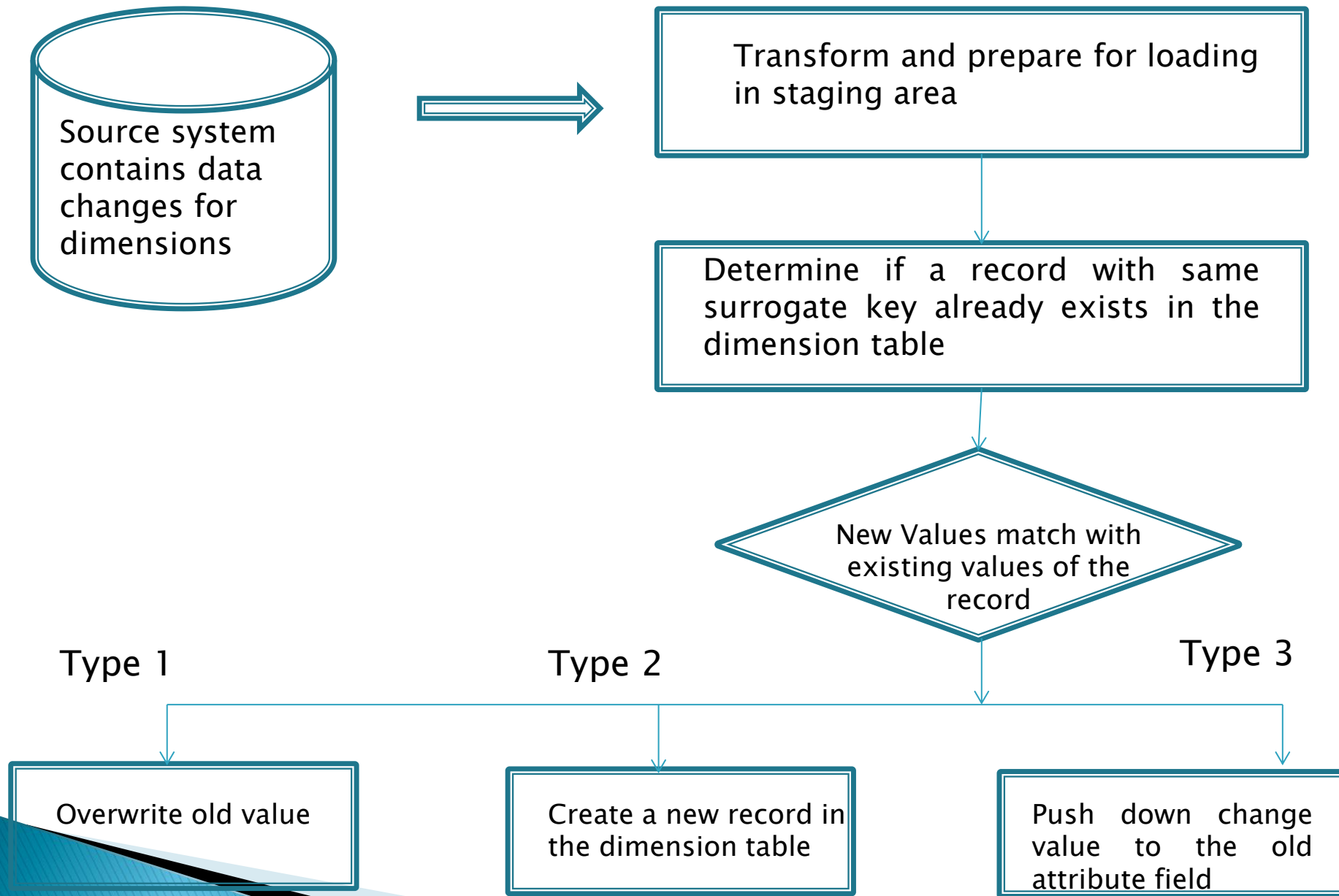
After the initial load, the data warehouse needs to be maintained and updated and this can be done by the following two methods:

Update—application of incremental changes in the data sources.

Refresh—complete reloads at specified intervals.



Loading fact table and the Dimension Tables



Data Quality

Accuracy:

There are many possible reasons for flawed or inaccurate data here. i.e. Having incorrect values of properties that could be human or computer errors.

Completeness:

For some reasons, incomplete data can occur, attributes of interest such as customer information for sales & transaction data may not always be available.

Consistency:

Incorrect data can also result from inconsistencies in naming convention or data codes, or from input field incoherent format. Duplicate tuples need cleaning of details, too.



Timeliness:

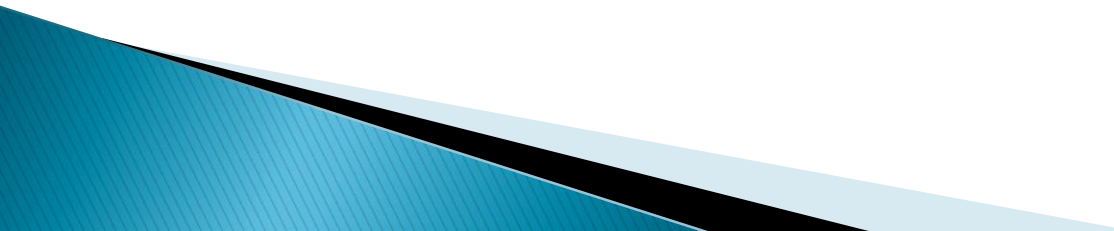
It also affects the quality of the data. At the end of the month, several sales representatives fail to file their sales records on time. There are also several corrections & adjustments which flow into after the end of the month. Data stored in the database are incomplete for a time after each month.

Believability:

It is reflective of how much users trust the data.

Interpretability:

It is a reflection of how easy the users can understand the data.



Category	ETL	ELT
Stands for	Extract, transform, and load	Extract, load, and transform
Process	Takes raw data, transforms it into a predetermined format, then loads it into the target data warehouse.	Takes raw data, loads it into the target data warehouse, then transforms it just before analytics.
Transformation and load locations	Transformation occurs in a secondary processing server.	Transformation takes place in the target data warehouse.
Data compatibility	Best with structured data.	Can handle structured, unstructured, and semi-structured data.
Speed	ETL is slower than ELT.	ELT is faster than ETL as it can use the internal resources of the data warehouse.
Costs	Can be time-consuming and costly to set up depending on ETL tools used.	More cost-efficient depending on the ELT infrastructure used.
Security	May require building custom applications to meet data protection requirements.	You can use built-in features of the target database to manage data protection.

Issues in Data Cleansing

Data cleaning is fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset. If data is incorrect, outcomes and algorithms are unreliable, even though they may look correct. When combining multiple data sources, there are many opportunities for data to be duplicated or mislabeled

- Remove duplicate or irrelevant observations

- Fix structural errors

- Filter unwanted outliers

- Handle missing data

